Informative and uninformative prestimulus cues at encoding benefit familiarity and source memory

Nicholas Yeh and Joshua D. Koen*

Department of Psychology, University of Notre Dame, IN

*Corresponding Author:

Joshua D. Koen

University of Notre Dame

Department of Psychology

Notre Dame, IN 46556

Email: jkoen@nd.edu

Abstract

Research using both neuroimaging and psychophysiology show that neural activity elicited by an informative prestimulus cue predicts subsequent memory. However, it remains unclear whether prestimulus encoding cues benefit subsequent memory performance. At encoding, participants made one of two semantic judgments on words preceded by an informative prestimulus cue that identified the upcoming semantic judgment, an uninformative prestimulus cue that signaled an upcoming trial but no information about the semantic judgment, or no cue. We observed that prestimulus cues affected multiple mnemonic processes. Specifically, dual process estimates of familiarity demonstrated a graded pattern with the informativeness of the prestimulus cues (i.e., informative > uninformative > no cues). Moreover, both informative and uninformative prestimulus cues enhanced subsequent source memory accuracy for the encoding task compared to the no cue condition. There was no evidence that prestimulus cues affected estimates of recollection derived from confidence-based receiver operating characteristic curves. These findings suggest that prestimulus cues can strengthen the processes that support successful memory encoding and benefit subsequent familiarity and source memory. The findings also suggest that prestimulus subsequent memory effects induced by prestimulus cues reflect the mobilization of multiple processes associated with successful memory encoding.

Keywords: Recognition Memory, Dual Process Theory, Encoding, Attention, Prestimulus Subsequent Memory Effects

Introduction

Efforts to understand how new experiences are encoded into memory have focused on the cognitive and neural processes elicited in the period immediately after the experience of an event. Most studies examining the neural correlates of successful encoding have used the subsequent memory procedure (Wagner et al., 1998) to investigate differences in neural activity elicited by stimuli that are later remembered or forgotten (Kim, 2011). However, there is now compelling evidence that ongoing neural activity in the seconds or so *prior to* the presentation of a stimulus predicts whether the stimulus will be subsequently remembered or forgotten (e.g., de Chastelaine & Rugg, 2015; Gruber & Otten, 2010; Otten et al., 2006, 2010; for review, see N. Cohen et al., 2015). Our aim in the present report is to expand on these neuroimaging findings to examine if prestimulus cues presented during encoding enhance subsequent recognition memory performance.

Research examining so-called prestimulus subsequent memory effects – differences in neural activity preceding the onset of a to-be-remembered stimulus as a function of subsequent memory – have concentrated on neural activity elicited by prestimulus cues at encoding that provided some information about the upcoming stimulus (Adcock et al., 2006; Addante et al., 2015; Bollinger et al., 2010; de Chastelaine & Rugg, 2015; Fellner et al., 2013; Galli et al., 2012; Gruber & Otten, 2010; Koen et al., 2018; Liu et al., 2020; Mackiewicz et al., 2006; Otten et al., 2006; Park & Rugg, 2010). For instance, Otten and colleagues (2006) recorded event-related potentials elicited by prestimulus cues that provided *valid* information about an upcoming trial, such as whether to make a semantic or orthographic judgment about the word. The authors used the Remember/Know procedure (Tulving, 1984) to dissociate the neural correlates of recollection and familiarity (Koen & Yonelinas, 2010; Yonelinas, 2001, 2002; for review, see Yonelinas et al., 2010; but see, Rotello et al., 2005), and observed negative going event-related potentials at frontal scalp electrodes elicited by cues preceding words that attracted subsequent Remember responses relative to words attracting subsequent Know and New responses (see also Otten et al., 2010). This finding was taken to suggest that prestimulus subsequent memory effects are associated with recollection. Additional studies have provided converging evidence for this proposal by showing prestimulus subsequent memory effects in other measures thought to reflect recollection, such as source memory (Koen et al., 2018; Liu et al., 2020) and associative memory (Addante et al., 2015). Moreover, prestimulus subsequent memory effects have been observed in the hippocampus (Addante et al., 2015; de Chastelaine & Rugg, 2015; Guderian et al., 2009; Liu et al., 2020; Merkow et al., 2014; Park & Rugg, 2010; Uncapher et al., 2011), a region that plays an important role in recolleciton (for reviews, see Diana et al., 2007; Eichenbaum et al., 2007; Yonelinas et al., 2010). The predominant conclusion reached by these studies is that neural activity elicited by prestimulus cues benefits encoding processes supportive of subsequent recollection (see also Salari & Rose, 2016).

The findings discussed above predict that providing informative prestimulus cues during encoding will benefit the processes supporting successful memory encoding and, in turn,

subsequent memory accuracy. However, this prediction has not been directly tested because the conditions necessary to test this hypothesis have not been included in much of the literature. For instance, most studies have included informative prestimulus cues (e.g., valid cues indicating a semantic or orthographic judgment about a word) but have not included a critical comparison condition that includes either an uninformative cue (e.g., neutral cues signaling a trial is upcoming but provides no information about performing a semantic or orthographic judgment about a word) or no prestimulus cue at all. The few studies that have included multiple types of cues have found that informative cues enhance memory compared to uninformative cues (Bollinger et al., 2010; 2011; although see Lin et al., 2017) and no prestimulus cues (Lin, et al., 2015). However, these studies were unable to parse out different mnemonic processes to fully understand the relationship between prestimulus cues and memory accuracy. Another related body of work examining the role of schemas and expectations on memory encoding have demonstrated that valid cues that accurately predict a feature of the upcoming stimulus (e.g., a natural or manmade object) facilitate familiarity whereas invalid cues enhance recollection (Kafkas, 2021; Kafkas & Montaldi, 2018; but see, Uncapher et al., 2011). The above studies also differ from prior neuroimaging work in that the prestimulus cues were not required to successfully complete the encoding task (but see, Galli et al., 2012; Gruber & Otten, 2010; Otten et al., 2006, Experiment 2). Taken together, while there is some evidence that prestimulus cues can benefit memory in certain conditions, the conditions in which these memory benefits arise and the mnemonic processes that prestimulus cues affect is still unclear.

Additionally, identifying when and under what circumstances prestimulus cues at encoding benefit subsequent memory performance will provide valuable theoretical insights about prestimulus subsequent memory effects. Of relevance to the present investigation are two hypotheses of prestimulus subsequent memory effects initially proposed by Otten and colleagues (2006; see also, Adcock et al., 2006; Park & Rugg, 2010). The first possible outcome is that informative and uninformative prestimulus cues relative to no prestimulus cues enhance memory accuracy with no memory differences emerging between the two prestimulus cues. This finding would align with a general attention hypothesis that prestimulus cues recruit general attentional resources that benefit memory encoding processes. A second potential outcome is that informative prestimulus cues, relative to both uninformative prestimulus cues and no prestimulus cues, enhances memory accuracy with no differences between uninformative and no prestimulus cues. If informative prestimulus cues proactively engage in task specific preparatory processes, this finding would align with a task set hypothesis. This hypothesis argues that prestimulus cues orient individuals to adopt an optimal task set through conscious, flexible, and strategic processes, which in turn benefits encoding of trials for which the task set is more effectively deployed in the prestimulus interval. It is important to point out that the task set and general attention hypotheses are not mutually exclusive, and thus it is possible that the processes by which prestimulus cues influence subsequent memory differ based on the mnemonic process being measured.

The present investigation examined whether informative and uninformative prestimulus cues during encoding benefited subsequent recognition memory relative to when no prestimulus cues were presented. We examined this issue by collating data from four pre-registered experiments that compared conditions with informative prestimulus cues, uninformative prestimulus cues, and no prestimulus cues (see Figure 1). We derived measures of recollection and familiarity from dual-process signal detection model fitted to receiver operator characteristic curves (Koen et al., 2017; Yonelinas, 1999; for reviews, see Yonelinas et al., 2010; Yonelinas & Parks, 2007) in each experiment, to examine whether prestimulus cues differentially affected recollection and familiarity. In addition, we included an objective measure of recollection (i.e., source memory for the semantic judgments made during encoding) in Experiments 2-4 to further explore whether prestimulus encoding cues affected subsequent memory performance. Given the similarity in the experimental designs and measures across all four experiments, we collated the data across relevant measures in each of the experiments and performed a linear mixed effects regression analysis. This approach helps to maximize the statistical power by pooling the data from all four experiments, while also accounting for error variance across experiments and participants. While this across-experiment analysis was not pre-registered, it closely follows the pre-registered analysis plan from each experiment.

Methods

Below, we focus on the general approach for the four experiments and the across-experiment analysis. The detailed methods and results for each individual experiment are included in the Supplemental Materials.

Ethics statement

These studies were approved by the Institutional Review Board of the University of Notre Dame. All participants provided informed consent before participation.

Preregistrations and Data Availability

We pre-registered the exclusion criteria, hypotheses, and analyses for each of the four experiments on the Open Science Framework. The preregistrations and data can be obtained from https://osf.io/ydmun/.

Participants

A total of 560 individuals were recruited to participant in one of four experiments. A total of 424 participants (234 females, 187 males, 3 preferred to not answer, M_{age} = 22.71, 18-30 years of age) contributed data to the across experiment analyses reported here. This sample comprised 64, 82, 130, and 148 participants from Experiments 1-4 respectively. Note that data from an additional 136 participants were collected and excluded from the analysis due to meeting one of the pre-registered exclusion criteria. Exclusionary criteria included failing to respond on 20% or more of either the encoding trials or test trials, low item memory (<.10), low source memory (<.05), and responding with the middle range of the confidence responses (i.e., 2-5 bins) on

fewer than 10% of all valid test trials (collapsed across all trial types). Note, Experiment 1 also had data excluded due to a fire alarm during the experimental session.

All participants were young adults fluent in English. Participants were recruited for in-person testing (Experiment 1) and online testing (Experiment 2) from the University of Notre Dame's online recruitment system (SONA). Additionally, some participants in Experiment 2, and all participants in Experiment 3-4, were recruited from online crowd sourcing platform Prolific (https://www.prolific.co/). Participants recruited from Prolific had an approval rating of 90% or greater (e.g., participant's performance metric for completing Prolific studies). This Prolific guideline is consistent with previous work (Peer et al., 2014). In addition, individuals who participated in one of the studies were ineligible to partake in the others. Participants for the four experiments were compensated either with course credit or $10.00 if they were recruited through the University of Notre Dame or through Prolific, respectively. For full breakdown of inclusionary, exclusionary, and behavioral performance criteria see Supplementary Materials.

Materials and Apparatus

The stimuli comprised of 384 nouns for Experiments 1-2 and 450 nouns for Experiments 3-4). The words were obtained from the MRC Psycholinguistic Database (Coltheart, 1981) with the following characteristics: 4-8 (M = 5.37, SD = 1.20) letters in length, ranged from 1-40 (M = 12.81, SD = 10.05) in written frequency (Kučera & Francis, 1967), and had concreteness ratings in the range of 500-662 (M = 582.94, SD = 31.83). An additional 45 words with similar characteristics were used to make practice encoding and test lists.

For Experiment 1, we generated a randomly generated stimulus list for each participant comprised of 256 words the encoding phase (128 words the informed and uninformed) with the remaining 128 words (new) being presented during the recognition test. For Experiment 2, the stimuli were the same as in Experiment 1 with the exception that 8 stimulus sets were pseudo randomly generated to match informed vs. no cue conditions (128 words in each condition) on the same word characteristics (length, frequency, and concreteness.) In Experiments 3 and, 66 additional words were added to the words from Experiments 1 and 2 and were used to create 4 stimuli sets with 300 words assigned to the study phase and 150 words assigned to be new items at test. The lists were created pseudo randomly matching the informed, uninformed, and no cue conditions (100 words in each condition) on the same word characteristics.

The stimuli for Experiment 1 was presented using PsychoPy v3.0.3 (Peirce, 2007; Peirce et al., 2019). The words were presented in Helvetica font at a height of 1.5 degrees of visual angle (using PsychoPy *degFlat* units). For Experiments 2-4, the stimuli were presented via Pavlovia.org using PsychoPy v2020.1.3 (Peirce, 2007; Peirce et al., 2019). Words were presented in white Arial font with a letter height of .05 (using *height* units) on a grey background.

Procedure

Across the four experiments the participants completed an incidental encoding task and recognition test. Experiments 2-4 included a source memory judgment for words attracting an old response. The recognition test occurred either immediately (Experiments 2-4) or 10 minutes (Experiment 1) following the encoding phase.

Encoding phase

During the incidental encoding phase, participants made one of two semantic judgments (shoebox, manmade) for each word. For the manmade judgments, participants responded 'yes' if the word denoted a manmade object, and 'no' if the word denoted a natural object. For the shoebox judgment, participants responded 'yes' if the object denoted by the word fit inside a shoebox, and 'no' otherwise. Participants were instructed to enter their keyboard responses (i.e., "j" yes and "k" for no) quickly and with high accuracy.

There were three different prestimulus cue conditions across the four experiments (informative, uninformative, or no cues). Informative cues signaled whether participants should make a manmade judgment (blue shape) or shoebox judgment (red shape) for upcoming words. Uninformative cues consisted of a white shape which signaled an upcoming trial without any semantic judgment information to be performed on the word. In this case, the semantic judgment was signaled simultaneously with the word (e.g., inside a blue or red box). The no prestimulus cue trials did not provide any information about the upcoming trial or the semantic judgment. The semantic judgment was signaled in the same way as in the uninformative cue trials.

In general, each trial was comprised of the presentation of a prestimulus cue (informative, uninformative, or no cue), a fixation cross, a word, and a final fixation cross (see Figure 1 for trial timing for each experiment). The response window included the time between the onset of a word and the final fixation cross. Participants completed a short practice phase before the critical encoding phase.

Experiment 1 presented informative and uninformative cues pseudorandomly and intermixed such that no more than 3 trials with the same cue condition or semantic judgment occurred consecutively. Experiment 2 presented informative and no prestimulus cues in a blocked design (2 informative, 2 no cue) with an ABBA/BAAB counterbalance scheme (A-informative, B- no cue). Experiment 3 presented all three cue conditions with a blocked design using an ABCCBA/BCAACB/CABBAC counterbalance scheme (A-informative, B-uninformative, C-no cue). Experiment 4 presented all three prestimulus cue conditions randomly intermixed in the same phase.

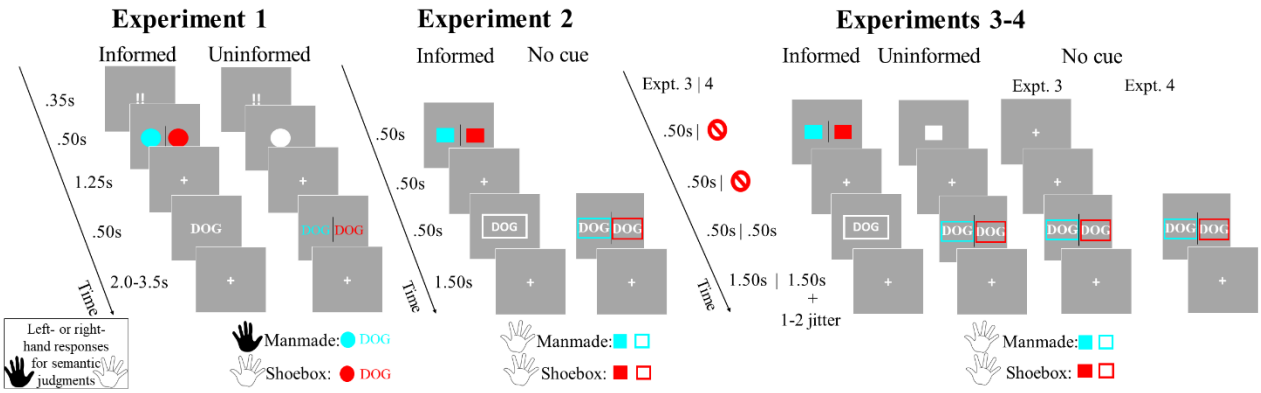

Figure 1. Depiction of the encoding task structure and trials timing for Experiments 1-4. Participants made semantic judgments (shoebox, manmade) about words preceded by either informative, uninformative, or no cues. Informative cues indicated ahead of time what semantic judgment to make for the words, with red and blue shapes signaling a shoebox and manmade judgment respectively. Uninformative cues, which were signaled by a white shape, indicated that a trial was upcoming but contained no information regarding the semantic judgment to make. Instead, the semantic judgment was signaled simultaneously with the word by either a red/blue font (Experiment 1) or red/blue box containing the word (Experiment 2-4). No cues contained no information about the upcoming trial or semantic judgment to make. For these trials, the semantic judgment was signaled in the same manner as the uninformative cue trials. The response window for the semantic judgments began with the word presentation and for the duration of the fixation cross following word presentation. Note, Experiment 4 included a 1-2s jitter following the response window time. Experiment 1 and 4 presented prestimulus cues intermixed. Experiment 2 and 3 presented prestimulus cues in individual blocks.

Recognition test

Following the encoding phase, participants completed a recognition memory test in which they identified studied words as "old" and words not presented in the encoding phase as "new" using a 6-point confidence scale (i.e., "1-sure new", "2- maybe new", "3-guess new", "4-guess old", "5-maybe old", "6-sure old") (Yonelinas & Parks, 2007). Participants were instructed to use the full range of confidence scale responses. In Experiments 2-4, following an old response (i.e., a 4, 5, or 6 confidence response), participants made a source memory decision by reporting their memory for the semantic judgment performed on the word during the encoding phase (i.e., shoebox, manmade, or don't know). Participants were instructed to select the shoebox or manmade judgment only if they had high confidence in their memory, otherwise they were instructed to respond with a "Don't Know" response. A "new" response (i.e., a 1, 2, or 3 confidence response) progressed the task to the next test item. In Experiment 1, the old/new responses were self-paced. In Experiments 2-4, both the old/new and source memory responses had a 10 sec time limit. There was a .5 second inter-trial interval of after the last response (item

or source memory response). Participants completed a practice test phase prior to the main recognition test to ensure they understood the instructions.

Dependent variables

Recollection and familiarity were estimated by fitting the dual-process signal detection model to receiver operating characteristics from confidence responses. Model fitting was carried out using the ROC Toolbox for Matlab (Koen et al., 2017) using the dual-process signal detection model. Our choice to use this model (Yonelinas, 1994, 1999) was driven by findings linking prestimulus subsequent memory effects to recollection (e.g., Otten et al., 2006, 2010; Park & Rugg, 2010) and based on prior work demonstrating that the parameters of the dual-process signal detection model behave in a theoretically meaningful way (Koen et al., 2013; Koen & Yonelinas, 2010, 2011; Yonelinas, 1999, 2001). The dual-process signal detection model estimates recollection as a probability ranging from 0 to 1 and familiarity as a discrimination index (d'); familiarity estimates were constrained to be positive (i.e., > 0). For Experiments 1 and 2, the model fitting routine estimated a total of 9 parameters: two estimates of recollection (informative and uninformative cues in Experiment 1; informative and no cue in Experiment 2), two estimates of familiarity, and five parameters modeling the location of response criterion. For Experiments 3 and 4, the model fitting routine estimated a total of 11 parameters: three estimates of recollection (informative, uninformative, no cue), three estimates of familiarity (informative, uninformative, no cue), and five parameters modeling the location of response criterion. The number of criterion parameters for confidence-based receiver operating characteristics is always N-1, where N is the number of confidence bins. The same criterion parameters were used for the informative, uninformative, and no cue conditions because there was only a single set of new items.

Additionally, source memory performance was computed for Experiments 2-4 accounting for the "guess rate" using a single high threshold model (Mattson et al., 2014; Snodgrass & Corwin, 1988). For correctly recognized old words, the proportion of correct source memory responses (Hit) and the proportion of "don't know" (DK) source responses were used to calculate source memory accuracy (i.e., probability of source recollection; pSR) with the following formula:

$$pSR = \frac{Hit - .5 * [1 - DK]}{1 - .5 * [1 - DK]}$$

Statistical Analyses

The statistical analyses comprised of a series of linear mixed effects regression models conducted on the dependent measures extracted from the four experiments. The linear mixed effects regression approach has several advantages over traditional ANOVA analyses, such as

modeling a nested structure (e.g., participants within experiments), handling missing observations (e.g., experiments not including all cue conditions or all memory measures), and appropriately accounting for random effects (e.g., participants varying in memory performance for the cue conditions across experiments). The mixed effects analyses were conducted in R (version 3.6.3) with the *mixed* function in the *afex* package (Singmann et al., 2021) and follow up post hoc tests being carried out with the *emmeans* package (Lenth, 2021). These models had a single fixed effect cue condition (informative, uninformative, no cue). There were also two random intercepts for participants and experiment, with participants nested within experiments. Specifically, the mixed effects model formula was:

$$Memory_{ij} = \beta_{00} + \beta_{01} cue\ condition_{ij} + \left(z_{Experiment_i}\ /\ u_{Participant_j}\right)$$

In the above equation, $Memory_{ji}$, represents the predicted memory measure for Participant *j* and Experiment *i*, β_{00} is the model intercept, the beta coefficient (β_{01}) represent the cue condition fixed effect, $z_{Experiment_i}\ /\ u_{Participant_j}$represents that Participant *j* is nested within Experiment *i*. A total of 4 mixed effects analyses were carried out on corrected hit rates, familiarity estimates, recollection estimates, and source memory (pSR). Degrees of freedom for the fixed effects were estimated with the Kenward-Roger method. We report effects sizes of partial-eta squared (η_p^2) for the main effects of cue condition and, where appropriate, Cohen's *d* for post-hoc comparisons along with the 95% confidence interval of the effect sizes (J. D. Cohen, 1988).

Results

The analyses reported below focus on the familiarity and recollection estimates of studied words derived from receiver operating characteristics curves and on estimates of source memory accuracy. The analysis of encoding reaction times and the corrected hit rates (i.e., hit rate minus false alarm rates) are reported in the Supplemental Material. Additionally, the conclusions we reached based on the analyses reported below were unchanged when we analyzed the data from each experiment individually using the recently proposed reverse Bonferroni method (Francis & Thunell, 2021). This method controls for Type 1 and 2 error rates by adjusting the alpha criterion level upwards ($\alpha = .05^{1/k}$; k = number of tests within a family of comparisons) when drawing conclusions that require consistent results across direct and conceptual replications (i.e., all results are significant in each experiment) to support a hypothesis.

Dual Process Signal Detection Model Parameter Estimates

Familiarity

The results from the linear mixed model conducted on familiarity estimates also demonstrated a significant effect of cue condition, $F(2, 703.90) = 20.27$, $p < .001$, $\eta_p^2 = .05$ [95% CI: .03, .09] (see Figure 2). Post-hoc contrasts demonstrated a graded effect across the three cue conditions like that observed for the corrected hit rates (i.e., informative > uninformative > no cue). Specifically, familiarity estimates were significantly higher for words preceded by informative, $t(705.87) = 6.37$, $p < .001$, $d = .48$ [95% CI: .33, .63], and uninformative prestimulus cues, $t(703.91) = 3.28$, $p = .001$, $d = .25$ [95% CI: .10, .40], relative to no prestimulus cues. Familiarity estimates were also higher for words preceded by informative prestimulus encoding cues relative to uninformative prestimulus encoding cues, $t(704.20) = 2.73$, $p = .007$, $d = .21$ [95% CI: .06, .35].

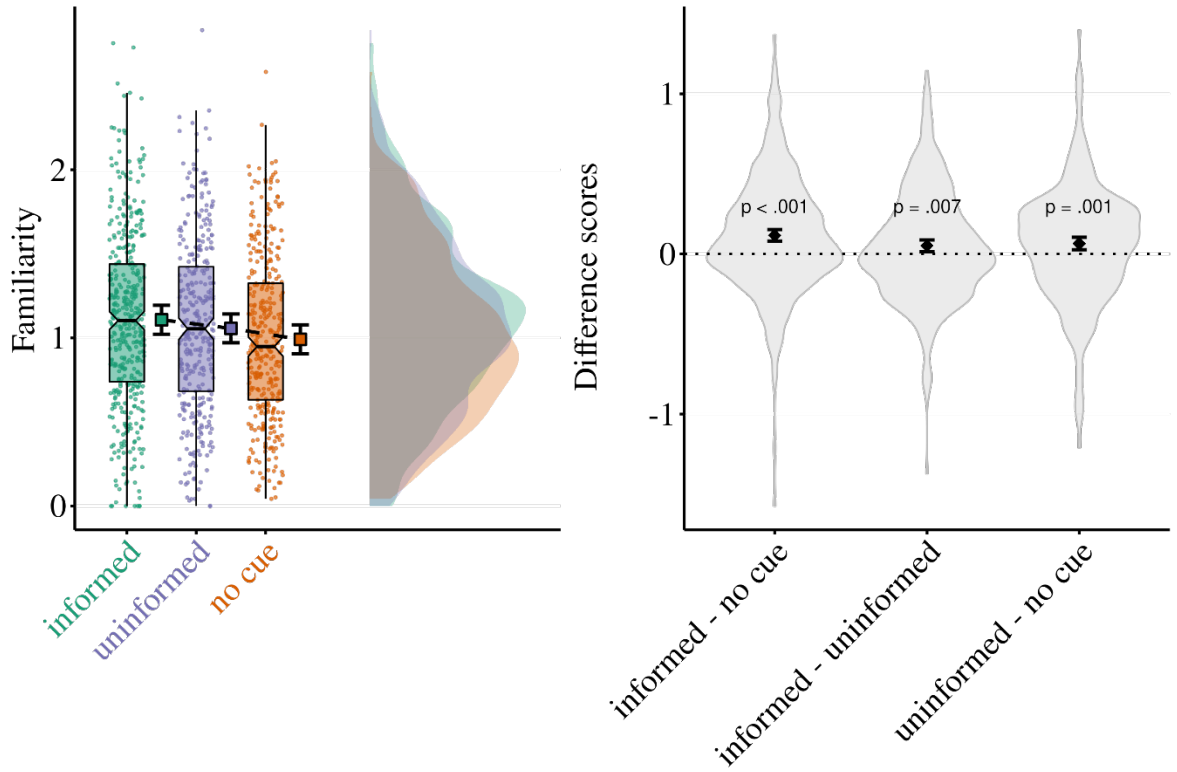

Figure 2. Results for the familiarity estimates. Left panel displays raincloud plots. The boxplots show the median, notches (95% confidence intervals), and the IQR with lower and upper whiskers (1.5 * IQR min and max). The means of the left panel reflect the model implied mean and 95% confidence interval. The right panel displays the violin plots for corrected hit rate difference scores between the three cued conditions along with the model implied mean differences and 95% confidence interval.

Recollection

There was no significant effect of cue condition on estimates of recollection derived from the dual process signal detection model, $F(2, 700.81) = 27.94$, $p < .001$, $\eta_p^2 = .01$ [95% CI: .00, .05] (see Figure 3).

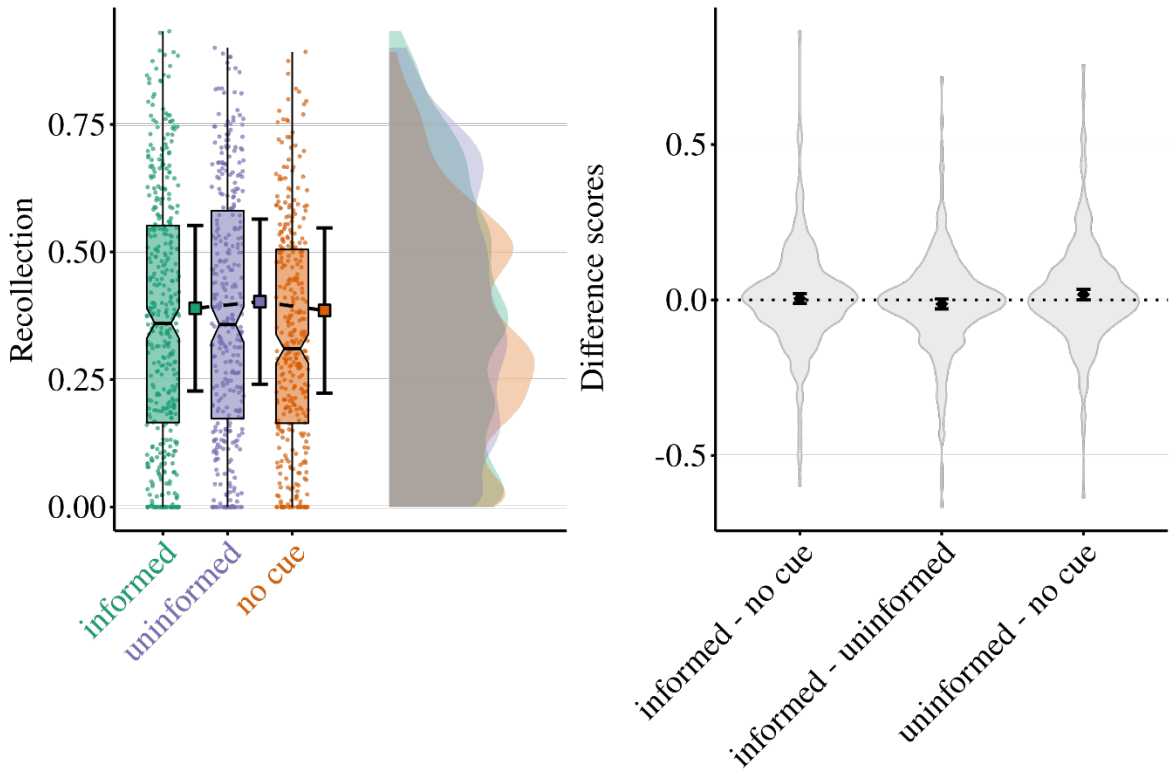

Figure 3. Results for the recollection estimates. Left panel displays raincloud plots. The boxplots show the median, notches (95% confidence intervals), and the IQR with lower and upper whiskers (1.5 * IQR min and max). The means of the left panel reflect the model implied mean and 95% confidence interval. The right panel displays the violin plots for corrected hit rate difference scores between the three cued conditions along with the model implied mean differences and 95% confidence interval.

Source Memory

The results from the linear mixed effects regression analysis also revealed a significant cue condition effect on source memory accuracy (i.e., pSR), $F(2, 637.77) = 6.72$, $p < .001$, $\eta_p^2 = .02$ [95% CI: .00, .05] (see Figure 4). This effect was driven by source memory performance being higher for words preceded by informative, $t(636.09) = 3.14$, $p = .002$, $d = .25$ [95% CI: .09, .41], and uninformative prestimulus cues, $t(638.62) = 3.11$, $p = .002$, $d = .25$ [95% CI: .09, .40], relative to the no prestimulus cue condition. The difference in source memory accuracy between informative and uninformative prestimulus cues was not significant, $t(638.62) = .27$, $p = .788$, $d = .02$ [95% CI: -.18, .13]. These results suggest that source memory for the semantic judgment was enhanced when words were preceded by both informative and uninformative prestimulus cues.

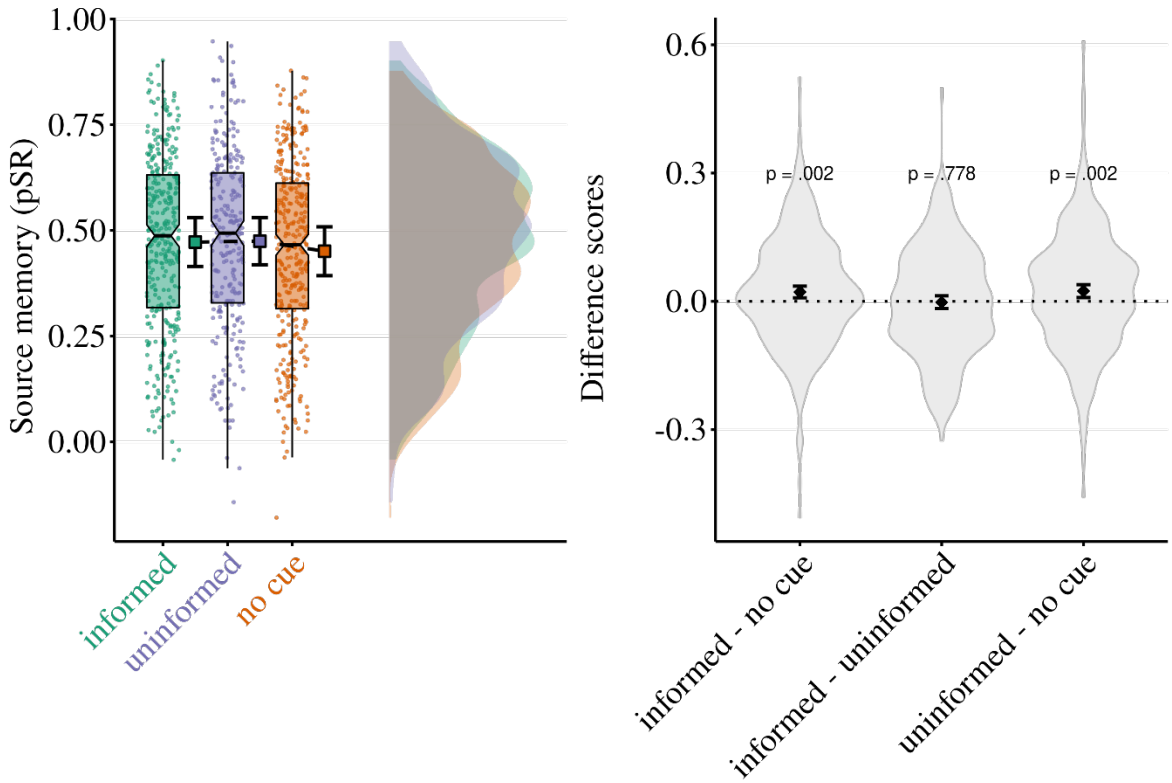

Figure 4. Results for source memory. Left panel displays raincloud plots. The boxplots show the median, notches (95% confidence intervals), and the IQR with lower and upper whiskers (1.5 * IQR min and max). The means of the left panel reflect the model implied mean and 95% confidence interval. The right panel displays the violin plots for corrected hit rate difference

scores between the three cued conditions along with the model implied mean differences and 95% confidence interval.

Discussion

The present study investigated whether prestimulus encoding cues enhanced subsequent memory performance. Across four experiments, participants encoded words while making one of two semantic judgments that were preceded by either informative cues signaling the semantic judgment to make, uninformative cues that only signaled an upcoming trial but not the semantic judgment, or were not preceded by a prestimulus cue. The informative cues signaled the semantic judgment 1.5 seconds prior to presentation of the word, whereas the semantic judgment for the uninformative and no cue conditions were signaled simultaneously with the onset of the word. Following the study phase, recognition memory was assessed using old/new confidence ratings and source memory for the semantic judgment at encoding for words attracting an old response. There were two primary findings from the across-experiment analysis. First, we demonstrated that familiarity estimates showed a graded effect with the informativeness of the prestimulus cues. Specifically, words encoded with an informative prestimulus cue had the highest familiarity estimates compared to both uninformative and no prestimulus cues, and words encoded with uninformative prestimulus cues had higher familiarity estimates compared to the no prestimulus cue condition. Second, although there was no effect of prestimulus cues on estimates of recollection derived from the dual process signal detection model, we observed that prestimulus informative and uninformative prestimulus cues benefited an objective measure of recollection (i.e., source memory) relative to the no cue condition. Source memory accuracy, however, did not differ between informative and uninformative prestimulus cues. Overall, prestimulus cues enhanced memory performance, but this pattern differed depending on the mnemonic process.

Prior EEG/ERP and fMRI studies (de Chastelaine & Rugg, 2015; Koen et al., 2018; Liu et al., 2020; Otten et al., 2006) have argued that so-called prestimulus subsequent memory effects are associated with encoding processes that support subsequent recollection. Our results are not fully consistent with this proposal for a couple of reasons. First, we only observed that prestimulus cues enhanced objective (source memory), but not subjective, measures of recollection. Second, and more importantly, we found that prestimulus cues also affected estimates of familiarity (for related findings, Kafkas, 2021; Kafkas & Montaldi, 2018). Prior work may have missed the relationship between prestimulus subsequent memory effects and familiarity because they have not clearly isolated recollection from familiarity. For example, previous findings have contrasted neural activity associated with subsequent Remember responses from that associated with a combination of subsequent Know and New responses (e.g., Otten et al., 2006). Although some work has, arguably, more clearly linked prestimulus subsequent memory effects to recollection (e.g., de Chastelaine & Rugg, 2015; Otten et al., 2010; Park & Rugg, 2010), the links to familiarity have not been thoroughly investigated. These

findings provide evidence the prestimulus cues are associated with both recollection and familiarity.

We argue that our results suggest a full theoretical understanding of EEG/ERP and fMRI prestimulus subsequent memory effects must account for different mnemonic processes, namely recollection and familiarity. As outlined in the Introduction, two prominent theories of prestimulus subsequent memory effects relevant to the current study are the task set and general attention hypotheses. The results from the analysis of familiarity estimates are most consistent with the task set hypothesis. This hypothesis proposes that prestimulus cues orient individuals to engage an optimal task set through conscious, flexible, and strategic processes. Adoption of these proactive processes are proposed to benefit encoding of trials for which the task set is more effectively deployed in the prestimulus interval (Otten et al., 2006, 2010). We argue the task set hypothesis predicts that more informative cues will increase the likelihood of adopting the appropriate task set, and thus will have the largest benefit to subsequent memory performance, which is consistent with the pattern of results we observed for familiarity estimates. However, familiarity showed a graded relationship with the prestimulus encoding cue condition (i.e., informative > uninformative > no cue; see Figure 2). A strict interpretation of this hypothesis, we argue, predicts that only informative cues would benefit subsequent memory performance. Thus, it is unclear if this finding is wholly consistent with the task set hypothesis. However, the task set and general attention hypotheses are not mutually exclusive. One way to reconcile the prestimulus subsequent memory findings with respect to our familiarity findings is to assume that the effects of activating a task set and general attention resources are additive. Given this assumption, one would expect a graded effect whereby informative prestimulus cues benefit memory encoding compared to uninformative prestimulus cues which, in turn, leads to better memory performance compared to trials with no prestimulus encoding cues. Future work is needed to examine the implications of these findings more fully.

Turning to the pattern of recollection findings, although no differences in subjective recollection estimates were observed, prestimulus cues did enhance an objective measure of recollection, namely source memory performance. In contrast to the observed graded familiarity findings, the pattern of source memory enhancement is more easily reconciled by prestimulus cues engaging general attention processes. From this perspective, both informative and uninformative prestimulus cues during encoding are predicted to equally recruit general attentional resources that benefit memory encoding processes (cf. DeCarlo, 2002). Our findings converge with this general attention prediction as source memory was enhanced for words preceded by both informative and uninformative prestimulus cues. Taken together, our source memory and familiarity findings described above suggest that a full account of prestimulus subsequent memory effects necessitates consideration of multiple encoding and mnemonic processes.

Another interesting aspect of the present findings is that prestimulus cues appeared to have a specific effect on objective measures of recollection but not on subjective measures

derived from confidence ratings and receiver operating characteristic curves. This was surprising, as these two measures are thought to capture the underlying mnemonic process (Yonelinas, 1999) and because prestimulus subsequent memory effects have been observed with both objective (Addante et al., 2015; de Chastelaine & Rugg, 2015; Ezzyat et al., 2017; Galli et al., 2012; Koen et al., 2018; Liu et al., 2020) and subjective (Galli et al., 2012, 2013; Otten et al., 2006, 2010; Park & Rugg, 2010) measures of recollection. The reason for this discrepancy is unclear. One possibility is that non-criterial recollection may have contributed to the diverging pattern of results for subjective and objective measures of recollection (Parks, 2007; Toth & Parks, 2006; Yonelinas & Jacoby, 1996). Another possibility is that the null effect on subjective measures of recollection resulted from mismodelling of the receiver operator characteristic curves (for a discussion of this issue, see Jang et al., 2012; Klauer & Kellen, 2015; Koen et al., 2013; Koen & Yonelinas, 2010, 2013; Spanton & Berry, 2020; Starns et al., 2012; for reviews of signal detection models of memory, see Wixted, 2007; Yonelinas & Parks, 2007). Additional research is needed to investigate this potential limitation and identify if prestimulus cues differentially affect objective and subjective measures of recollection. Critically, if our finding that prestimulus cues at encoding benefit objective but not subjective measures of recollection holds, then this would provide an important constraint on theories of prestimulus subsequent memory effects.

The present work demonstrates that prestimulus encoding cues affect multiple encoding processes that are differentially associated with subsequent recollection and familiarity. More specifically, a graded familiarity memory pattern emerged when memory was assessed with dual-process signal detection model. Additionally, both prestimulus cues enhanced source memory performance while leaving estimates of recollection unaffected. Future research can expand upon these findings to determine how prestimulus cues affect other types of memory (e.g., priming, Düzel et al., 2005). Furthermore, future work using electrophysiological and neuroimaging techniques may uncover multiple processes that contribute to the influence of prestimulus cues on memory encoding through a systematic comparison of different types of cues.

Acknowledgements

The results reported here were presented at the 2020 Psychonomic Society Virtual Meeting. We would like to thank Dr. Jessica D. Payne, Dr. Michael D. Rugg, Morgan Munsen, and Rachelle Pichot for providing comments on a previous version of this manuscript. The authors report no conflict of interest.

Joshua D. Koen was supported by grant R56AG068149 from the National Institute on Aging during the preparation of this manuscript. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

References

Adcock, R. A., Thangavel, A., Whitfield-Gabrieli, S., Knutson, B., & Gabrieli, J. D. E. (2006). Reward-Motivated Learning: Mesolimbic Activation Precedes Memory Formation. *Neuron*, *50*(3), 507–517. https://doi.org/10.1016/j.neuron.2006.03.036

Addante, R. J., de Chastelaine, M., & Rugg, M. D. (2015). Pre-stimulus neural activity predicts successful encoding of inter-item associations. *NeuroImage*, *105*, 21–31. https://doi.org/10.1016/j.neuroimage.2014.10.046

Bollinger, J., Rubens, M. T., Zanto, T. P., & Gazzaley, A. (2010). Expectation-Driven Changes in Cortical Functional Connectivity Influence Working Memory and Long-Term Memory Performance. *Journal of Neuroscience*, *30*(43), 14399–14410. https://doi.org/10.1523/jneurosci.1547-10.2010

Cohen, J. D. (1988). *Statistical power analysis for the social sciences* (2nd ed., Vol. 2nd). Lawrence Erlbaum.

Cohen, N., Pell, L., Edelson, M. G., Ben-Yakov, A., Pine, A., & Dudai, Y. (2015). Peri-encoding predictors of memory encoding and consolidation. *Neuroscience & Biobehavioral Reviews*, *50*, 128–142. https://doi.org/10.1016/j.neubiorev.2014.11.002

de Chastelaine, M., & Rugg, M. D. (2015). The effects of study task on prestimulus subsequent memory effects in the hippocampus: Pre-Stimulus Activity and Depth of Encoding. *Hippocampus*, *25*(11), 1217–1223. https://doi.org/10.1002/hipo.22489

DeCarlo, L. T. (2002). Signal detection theory with finite mixture distributions: Theoretical developments with applications to recognition memory. *Psychological Review*, *109*(4), 710–721. https://doi.org/10.1037//0033-295x.109.4.710

Diana, R. A., Yonelinas, A. P., & Ranganath, C. (2007). Imaging recollection and familiarity in the medial temporal lobe: A three-component model. *Trends in Cognitive Sciences*, *11*(9), 379–386. https://doi.org/10.1016/j.tics.2007.08.001

Düzel, E., Richardson-Klavehn, A., Neufang, M., Schott, B. H., Scholz, M., & Heinze, H.-J. (2005). Early, partly anticipatory, neural oscillations during identification set the stage for priming. *NeuroImage*, *25*(3), 690–700. https://doi.org/10.1016/j.neuroimage.2004.11.038

Eichenbaum, H., Yonelinas, A. P., & Ranganath, C. (2007). The Medial Temporal Lobe and Recognition Memory. *Annual Review of Neuroscience*, *30*(1), 123–152. https://doi.org/10.1146/annurev.neuro.30.051606.094328

Ezzyat, Y., Kragel, J. E., Burke, J. F., Levy, D. F., Lyalenko, A., Wanda, P., O'Sullivan, L., Hurley, K. B., Busygin, S., Pedisich, I., Sperling, M. R., Worrell, G. A., Kucewicz, M. T., Davis, K. A., Lucas, T. H., Inman, C. S., Lega, B. C., Jobst, B. C., Sheth, S. A., … Kahana, M. J. (2017). Direct Brain Stimulation Modulates Encoding States and Memory Performance in Humans. *Curr. Biol.*, *27*(9), 1251–1258. https://doi.org/10.1016/j.cub.2017.03.028

Fellner, M.-C., Bäuml, K.-H. T., & Hanslmayr, S. (2013). Brain oscillatory subsequent memory effects differ in power and long-range synchronization between semantic and survival processing. *NeuroImage*, *79*, 361–370. https://doi.org/10.1016/j.neuroimage.2013.04.121

Francis, G., & Thunell, E. (2021). Reversing Bonferroni. *Psychonomic Bulletin & Review*, *28*(3), 788–794. https://doi.org/10/gjf8b7

Galli, G., Choy, T. L., & Otten, L. J. (2012). Prestimulus brain activity predicts primacy in list learning. *Cognitive Neuroscience*, *3*(3–4), 160–167. https://doi.org/10.1080/17588928.2012.670105

Galli, G., Gebert, A. D., & Otten, L. J. (2013). Available processing resources influence encoding-related brain activity before an event. *Cortex*, *49*(8), 2239–2248. https://doi.org/10.1016/j.cortex.2012.10.011

Gruber, M. J., & Otten, L. J. (2010). Voluntary Control over Prestimulus Activity Related to Encoding. *Journal of Neuroscience*, *30*(29), 9793–9800. https://doi.org/10.1523/jneurosci.0915-10.2010

Guderian, S., Schott, B. H., Richardson-Klavehn, A., & Duzel, E. (2009). Medial temporal theta state before an event predicts episodic encoding success in humans. *Proceedings of the National Academy of Sciences*, *106*(13), 5365–5370. https://doi.org/10.1073/pnas.0900289106

Jang, Y., Mickes, L., & Wixted, J. T. (2012). Three tests and three corrections: Comment on Koen and Yonelinas (2010). *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *38*(2), 513–523. https://doi.org/10.1037/a0025880

Kafkas, A. (2021). Encoding-linked pupil response is modulated by expected and unexpected novelty: Implications for memory formation and neurotransmission. *Neurobiology of Learning and Memory*, *180*, 107412. https://doi.org/10/gkzm9h

Kafkas, A., & Montaldi, D. (2018). Expectation affects learning and modulates memory experience at retrieval. *Cognition*, *180*, 123–134. https://doi.org/10/gff8h5

Kim, H. (2011). Neural activity that predicts subsequent memory and forgetting: A meta-analysis of 74 fMRI studies. *NeuroImage*, *54*(3), 2446–2461. https://doi.org/10.1016/j.neuroimage.2010.09.045

Klauer, K. C., & Kellen, D. (2015). The flexibility of models of recognition memory: The case of confidence ratings. *J. Math. Psychol.*, *67*, 8–25. https://doi.org/10.1016/j.jmp.2015.05.002

Koen, J. D., Aly, M., Wang, W.-C., & Yonelinas, A. P. (2013). Examining the causes of memory strength variability: Recollection, attention failure, or encoding variability? *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *39*(6), 1726–1741. https://doi.org/10.1037/a0033671

Koen, J. D., Barrett, F. S., Harlow, I. M., & Yonelinas, A. P. (2017). The ROC Toolbox: A toolbox for analyzing receiver-operating characteristics derived from confidence ratings. *Behavior Research Methods*, *49*(4), 1399–1406. https://doi.org/10.3758/s13428-016-0796-z

Koen, J. D., Horne, E. D., Hauck, N., & Rugg, M. D. (2018). Age-related differences in prestimulus subsequent memory effects assessed with event-related potentials. *Journal of Cognitive Neuroscience*, *30*(6), 829–850. https://doi.org/10.1162/jocn_a_01249

Koen, J. D., & Yonelinas, A. P. (2010). Memory variability is due to the contribution of recollection and familiarity, not to encoding variability. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *36*(6), 1536–1542. https://doi.org/10.1037/a0020448

Koen, J. D., & Yonelinas, A. P. (2011). From humans to rats and back again: Bridging the divide between human and animal studies of recognition memory with receiver operating characteristics. *Learning & Memory*, *18*(8), 519–522. https://doi.org/10.1101/lm.2214511

Koen, J. D., & Yonelinas, A. P. (2013). Still no evidence for the encoding variability hypothesis: A reply to Jang, Mickes, and Wixted (2012) and Starns, Rotello, and Ratcliff (2012). *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *39*(1), 304–312. https://doi.org/10.1037/a0028462

Lenth, R. V. (2021). *emmeans: Estimated marginal means, aka least-squares means* [Manual]. https://CRAN.R-project.org/package=emmeans

Liu, E. S., Koen, J. D., & Rugg, M. D. (2020). Effects of age on prestimulus neural activity predictive of successful memory encoding: An fMRI study. *Cerebral Cortex*. https://doi.org/10.1093/cercor/bhaa265

Mackiewicz, K. L., Sarinopoulos, I., Cleven, K. L., & Nitschke, J. B. (2006). The effect of anticipation and the specificity of sex differences for amygdala and hippocampus function in emotional memory. *Proceedings of the National Academy of Sciences*, *103*(38), 14200–14205. https://doi.org/10.1073/pnas.0601648103

Mattson, J. T., Wang, T. H., de Chastelaine, M., & Rugg, M. D. (2014). Effects of Age on Negative Subsequent Memory Effects Associated with the Encoding of Item and Item-Context Information. *Cerebral Cortex*, *24*(12), 3322–3333. https://doi.org/10.1093/cercor/bht193

Merkow, M. B., Burke, J. F., Stein, J. M., & Kahana, M. J. (2014). Prestimulus theta in the human hippocampus predicts subsequent recognition but not recall: Item Recognition Versus Free Recall. *Hippocampus*, *24*(12), 1562–1569. https://doi.org/10.1002/hipo.22335

Otten, L. J., Quayle, A. H., Akram, S., Ditewig, T. A., & Rugg, M. D. (2006). Brain activity before an event predicts later recollection. *Nature Neuroscience*, *9*(4), 489–491. https://doi.org/10.1038/nn1663

Otten, L. J., Quayle, A. H., & Puvaneswaran, B. (2010). Prestimulus Subsequent Memory Effects for Auditory and Visual Events. *Journal of Cognitive Neuroscience*, *22*(6), 1212–1223. https://doi.org/10.1162/jocn.2009.21298

Park, H., & Rugg, M. D. (2010). Prestimulus hippocampal activity predicts later recollection. *Hippocampus*, *20*, 24–28. https://doi.org/10.1002/hipo.20663

Parks, C. M. (2007). The role of noncriterial recollection in estimating recollection and familiarity. *Journal of Memory and Language*, *57*(1), 81–100. https://doi.org/10.1016/j.jml.2007.03.003

Peer, E., Vosgerau, J., & Acquisti, A. (2014). Reputation as a sufficient condition for data quality on Amazon Mechanical Turk. *Behavior Research Methods*, *46*(4), 1023–1031. https://doi.org/10.3758/s13428-013-0434-y

Peirce, J. W. (2007). PsychoPy—Psychophysics software in Python. *Journal of Neuroscience Methods*, *162*(1–2), 8–13. https://doi.org/10.1016/j.jneumeth.2006.11.017

Peirce, J. W., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., Kastman, E., & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods*, *51*(1), 195–203. https://doi.org/10.3758/s13428-018-01193-y

Rotello, C. M., Macmillan, N. A., Reeder, J. A., & Wong, M. (2005). The remember response: Subject to bias, graded, and not a process-pure indicator of recollection. *Psychonomic Bulletin & Review*, *12*(5), 865–873. https://doi.org/10.3758/bf03196778

Salari, N., & Rose, M. (2016). Dissociation of the functional relevance of different pre-stimulus oscillatory activity for memory formation. *NeuroImage*, *125*, 1013–1021. https://doi.org/10.1016/j.neuroimage.2015.10.037

Singmann, H., Bolker, B., Westfall, J., Aust, F., & Ben-Shachar, M. S. (2021). *afex: Analysis of factorial experiments* [Manual]. https://CRAN.R-project.org/package=afex

Snodgrass, J. G., & Corwin, J. (1988). Pragmatics of Measuring Recognition Memory: Applications to Dementia and Amnesia. *Journal of Experimental Psychology: General*, *117*(1), 34–50. https://doi.org/10.1037/0096-3445.117.1.34

Spanton, R. W., & Berry, C. J. (2020). The unequal variance signal-detection model of recognition memory: Investigating the encoding variability hypothesis. *Quarterly Journal of Experimental Psychology*, *73*(8), 1242–1260. https://doi.org/10.1177/1747021820906117

Starns, J. J., Rotello, C. M., & Ratcliff, R. (2012). Mixing strong and weak targets provides no evidence against the unequal-variance explanation of z̨ROC slope: A comment on Koen and Yonelinas (2010). *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *38*(3), 793–801. https://doi.org/10.1037/a0027040

Toth, J. P., & Parks, C. M. (2006). Effects of age on estimated familiarity in the process dissociation procedure: The role of noncriterial recollection. *Memory & Cognition*, *34*(3), 527–537. https://doi.org/10.3758/bf03193576

Tulving, E. (1984). Précis of Elements of episodic memory. *Behavioral and Brain Sciences*, *7*(2), 223–238. https://doi.org/10.1017/s0140525x0004440x

Uncapher, M. R., Hutchinson, J. B., & Wagner, A. D. (2011). Dissociable Effects of Top-Down and Bottom-Up Attention during Episodic Encoding. *Journal of Neuroscience*, *31*(35), 12613–12628. https://doi.org/10.1523/jneurosci.0152-11.2011

Wagner, A. D., Schacter, D. L., Rotte, M., Koutstaal, W., Maril, A., Dale, A. M., Rosen, B. R., & Buckner, R. L. (1998). Building memories: Remembering and forgetting of verbal experiences as predicted by brain activity. *Science*, *281*(5380), 1188–1191. https://doi.org/10.1126/science.281.5380.1188

Wixted, J. T. (2007). Dual-process theory and signal-detection theory of recognition memory. *Psychological Review*, *114*(1), 152–176. https://doi.org/10.1037/0033-295x.114.1.152

Yonelinas, A. P. (1994). Receiver-Operating Characteristics in Recognition Memory: Evidence for a Dual-Process Model. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *20*(6), 1341–1354. https://doi.org/10.1037/0278-7393.20.6.1341

Yonelinas, A. P. (1999). The Contribution of Recollection and Familiarity to Recognition and Source-Memory Judgments: A Formal Dual-Process Model and an Analysis of Receiver Operating Characteristics. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *25*(6), 1415–1434. https://doi.org/10.1037/0278-7393.25.6.1415

Yonelinas, A. P. (2001). Consciousness, control, and confidence: The 3 Cs of recognition memory. *Journal of Experimental Psychology: General*, *130*(3), 361–379. https://doi.org/10.1037//0096-3445.130.3.361

Yonelinas, A. P. (2002). The Nature of Recollection and Familiarity: A Review of 30 Years of Research. *Journal of Memory and Language*, *46*(3), 441–517. https://doi.org/10.1006/jmla.2002.2864

Yonelinas, A. P., Aly, M., Wang, W.-C., & Koen, J. D. (2010). Recollection and familiarity: Examining controversial assumptions and new directions. *Hippocampus*, *20*(11), 1178–1194. https://doi.org/10.1002/hipo.20864

Yonelinas, A. P., & Jacoby, L. L. (1996). Noncriterial Recollection: Familiarity as Automatic, Irrelevant Recollection. *Consciousness and Cognition*, *5*(1–2), 131–141. https://doi.org/10.1006/ccog.1996.0008

Yonelinas, A. P., & Parks, C. M. (2007). Receiver operating characteristics (ROCs) in recognition memory: A review. *Psychological Bulletin*, *133*(5), 800–832. https://doi.org/10.1037/0033-2909.133.5.800